

NORWEGIAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

IT3010 - INDIVIDUAL REFLECTION REPORT

RESEARCH-BASED INNOVATION METHODOLOGIES IN COMPUTER AND INFORMATION SCIENCE

Assignment 6 - Report

Group members

Jens Christian Valen LEYNSE

Group number: 13

September 1, 2026 (Spring 2020)

<https://www.ntnu.edu/studies/courses/IT3010>

Introduction

A peer review is a process where academics unknown to an article's authors assess its suitability and quality before a decision is taken on whether to publish it in the journal [1]. The reason this topic was chosen is because peer reviews are an integral part of the scholarly publishing ecosystem, offering both advantages and limitations worth understanding for future benefits and insight. Moreover, since evaluating empirical research is a key objective of the IT3010 course, this topic holds relevance to the course's aims. When implemented in a well thought out manner, peer reviews create conversations between authors and reviewers, provide authors with context for reviewer comments, and ultimately improve the final product [2].

The motivation behind this topic arises from the intriguing challenge posed by the third assignment of the course, which focused on individual peer reviews. This assignment sparked a deeper interest in the evaluation of empirical research. Comparing the paper being reviewed to our own group project, we observed a distinction in how effectively the reviewed paper addressed the given questions. This process enabled us to detect errors in our own paper while evaluating others, facilitating mutual feedback between the evaluated group and ourselves. Additionally, it prompted self-reflection and improvements in our own paper. How peer reviews managed to give this many benefits is the reason for selecting this topic for the report.

Regarding the debates surrounding the topic of peer reviewing, numerous studies have explored its usage and provided detailed descriptions of various practical applications. However, this report will primarily focus on relatable aspects of peer reviewing based on personal experience. The main motivation for delving deeper into this subject stemmed from a sense of inexperience, prompting curiosity about how other researchers perceive the peer reviewing process. The chosen research for this report prioritizes the examination of limitations over benefits, as these limitations had the most effect on the personal experience being discussed.

Theoretical background

Christine L. Ferguson's study on open peer review examines the removal of anonymity in contrast to the conventional single or double blind peer review process [2]. Anonymity in peer review ensures that all reviews are of equal importance, without any one review affecting the process more than the others. This approach has the advantage of enabling reviewers to express critical perspectives without fear of repercussion, although it also carries risk of leading to petty or overly harsh criticisms. Given the study's focus, it provided valuable insights into the limitations and benefits associated with traditional peer reviews, aligning with the underlying motivation of this report.

Arham Arshad et al. study, titled "Towards a More Structured Peer Review Process with Empirical Standards," examines the inadequacies of standard empirical reviews and categorizes scholarly peer reviews as ineffective, unreliable, and prejudiced [3]. To address these issues, the paper employs a standardized review checklist in an effort to demonstrate how it enables more structured peer reviews, which in turn should improve review quality, reliability, thoroughness, and readability [3]. The inclusion of assessment criteria in the form of a checklist-like template for each section of the paper provided valuable insights into the necessary inclusions that should be incorporated. As a result, it led to an improved end product thanks to how it addressed the initial challenge of the review, namely the uncertainty around the evaluation criteria.

Sandra P. Thomas's study, titled "The conceptual Debates and Empirical Evidence About the Peer Review Process for Scholarly Journals", examines the debates surrounding peer reviews, highlighting its flaws, strengths and associated implications[4]. Within these discussions the study identifies certain "myths" pertaining to the peer review process. These myths include the assumption that all scientists are peers of one another, genius will be recognised, and reviewers act in the interest of science as a whole[4]. These myths raise concerns about the integrity and reliability of the peer reviewing process, and are important to keep in mind when discussing the limitations of empirical peer reviews.

Paul A. M. Van Lange's study, titled "Why Authors Believe That Reviewers Stress Limiting Aspects of Manuscripts: The SLAM Effect in Peer Review" examines the tendency of reviewers to place significant emphasis on imperfections and limitations, often neglecting the innovative aspects of a manuscript [5]. In the context of the peer reviewing assignment's objective to provide valuable feedback to authors, this study

becomes relevant as it sheds light on a self-perceived notion that feedback primarily focuses on improving the paper by highlighting areas for enhancement, and in a less sense pay attention to discuss its positive aspects.

Experience report

During the semester project, a couple of notable experiences emerged while conducting the peer review. The assessment criteria were categorized into different criteria for each section of the paper. This approach parallels the use of checklists advocated by Arham Arshad et al, who highlight the advantage of providing reviewers with specific attribute lists that align closely with community expectations, enabling them to inform their reviews effectively [3]. One significant benefit of this method is its explicit statement of evaluation criteria, addressing the challenge posed by inexperience in performing the peer review process and expediting review time management by giving a clear view of the highest-value criteria to assess.

Inexperience also posed limitations on the review process, influencing the mindset of the report. As discussed by Paul A. M. Van Lange, the tendency to overly emphasize criteria imperfections is not uncommon[5]. In retrospect, when reviewing a high-standard paper, it became apparent that some comments were made to suggest minor improvements that may not have been crucial to address. An improvement to this would involve highlighting what worked well, rather than nitpicking on insignificant issues.

Another intriguing aspect of the process was the use of anonymous reviewers. On the positive side, this allowed for unbiased reviews, particularly beneficial since many reviewers lacked experience, enabling them to express their thoughts openly. As Christine L. Ferguson mentioned, a downside to this could be how the openness allows for little personal restrictions as to what the reviewer comment in the review [2]. This was remedied well by including the reviewers' confidence level. This addition lead to an easier time making appropriate choices when further enhancing the research draft being worked on by our group.

The final aspect of the experience relates back to the inexperience of the reviewer. The limited understanding of the reviewing process prompted further contemplation on how other peer reviewers perceived their own experiences. Sandra P. Thomas's study on the empirical peer reviewing process discusses the selection of "peers" and the loosely defined nature of this term, particularly when considering the open selection of possible topics for group projects [4]. In my own experience, the paper under review aimed to be easily comprehensible; however, improving it required a deeper understanding that surpassed the narrative. Therefore, being peers of the group entailed having at least a basic understanding of the topic to effectively contribute to its improvement.

Discussion

During the review process, certain limitations were encountered, prompting further contemplation on how other researchers have approached and perceived these challenges. The initial inexperience in evaluating a peers' group project was addressed by utilizing assessment criteria, similar to the approach discussed in Arham Arshad et al. study, to guide the evaluation process [3]. This method proved valuable in navigating the review, considering specific attributes outlined in the criteria. However, as the study debates the theoretical options for how to go about the structured approach, it doesn't address a physical example to compare to. The inexperience also influenced the mindset during the review, leading to a tendency to focus on minor imperfections rather than highlighting the strengths of the project. This tendency aligns with findings from Paul A. M. Van Lange's study, which suggest that reviewers often prioritize making changes over understanding the author's intent behind the content [5]. Furthermore, the data from the study finds that the evaluation of own and others' work to some extent are socially defined.

Another notable aspect of the review process was the use of anonymous reviewers. This approach allowed for unbiased feedback, a key advantage of peer reviews. Additionally, the importance of considering the reviewers' confidence levels when interpreting the feedback emerged as a valuable insight. The study of open peer reviews concluded that there wasn't a clear answer to how beneficial anonymity was to open reviews,

as the impact on the reviewers bias was uncertain [2]. Although the impact on the project itself may have been relatively minor, this aspect provided more insights of one of the advantages of peer reviews, further enriching the understanding of the peer review process.

Lastly, the evaluation of the collective level of the peers involved in the process, based on the reflections of the limitation of personal understandings indicates that being a peer reviewer requires a basic understanding of the topic to provide meaningful improvements. Based on the study with the conceptual debates, although the scope of the study was small, the results showed that more training for the reviewer would be of beneficial to the process [4].

References

- [1] Oates, B. J. (2006). Researching information systems and computing. Reviewing the literature, 74-76.
- [2] Ferguson, C. L. (2021, January). Open peer review.
- [3] Arshad, A., Ghaleb, T. A., Ralph, P. (2021, June). Towards a More Structured Peer Review Process with Empirical Standards.
- [4] Thomas, S. P. (2011, May-June). Conceptual Debates and Empirical Evidence About the Peer Review Process for Scholarly Journals.
- [5] Van Lange, P. A. M. (1999). Why Authors Believe That Reviewers Stress Limiting Aspects of Manuscripts: The SLAM Effect in Peer Review.